DAY 30 | POST-READ

LLM Evaluation and QA: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Golden dataset	Verified, representative, versioned, maintained. Fields: id, question, expected, relevant ids, tags, must include and must not include.
Prompt regression	Rerun the golden set for every prompt, model or retrieval change; compare per item and overall.
Deterministic checks	Phrases, JSON validity, length. Fast; run on every commit.
Model-graded checks	Faithfulness, tone, correctness rubrics. Slower; nightly or before release.
Context precision	Relevant retrieved / retrieved. Rank-aware versions reward good ordering.
Context recall	Needed evidence found / needed evidence.
BLEU	n-gram overlap with brevity penalty. Good for translation, weak for open answers.
Quality gate	A script with thresholds that fails the pipeline on regressions.

Part B: Going beyond today

1. Baselines from main

Compare a pull request's scores with the main branch's latest scores, not with fixed numbers only. A drop of more than a tolerance (for example 2 points) fails the gate even if the absolute score is still high.

2. Non-determinism

Model outputs vary between runs. Use temperature 0 for tests, run important items several times, and set thresholds with a little slack rather than demanding perfection.

3. Cost-aware evaluation

Run cheap deterministic checks on every push, a small golden subset with model judges on pull requests, and the full suite nightly. Track cost per evaluation run.

4. Ready-made tools

Frameworks such as promptfoo, DeepEval, Ragas and the OpenAI Evals project help define datasets, metrics and CI integrations. Concepts from today carry over directly; check each tool's

current documentation.

5. Keep the golden set alive

Add every production failure that users report as a new golden item, with an owner and date. The set grows to reflect reality.

Part C: Check yourself

- What makes a dataset golden rather than just a test set?
- Why might reducing k from 3 to 1 hurt quality?
- What is the difference between context precision and context recall?
- Why can BLEU mislead for RAG answers?
- How does a quality gate stop a bad release?

Part D: Practice ideas for the weekend

- Add 10 golden items for your capstone domain with must include and must not include phrases.
- Add a baseline comparison to the quality gate.
- Wire the quality gate into the Azure pipeline from Day 19 as a stage before deployment.

Coming up on Day 31: LLMOps: the LLM lifecycle, what breaks in production, escalations and emotional intelligence for FDEs.

Yoi shuumatsu wo (良い週末を): have a great weekend